

15. Let $i = \sqrt{-1}$. Find the exact value of $|\sqrt{6} + i\sqrt{8}|$. Write your answer in simplified radical form.
16. Tom is a gambler. He has a pair of fair, standard cubical dice. He selects two different numbers in advance from the face numbers 1, 2, 3, 4, 5, and 6. He then rolls the pair of dice. If both numbers showing match his two selections, he wins \$108. If exactly one number showing matches one of his two selections, he neither wins nor loses. If neither number showing matches either of his two selections, he loses \$54. Find Tom's mathematical expectation each time he rolls the pair of dice. Express your answer by including the word "lose" or the word "win." For example, your answer might be "lose \$2" or might be "win \$7."
17. Let D be the point that is $\frac{1}{3}$ of the way from B to A for equilateral triangle $\triangle ABC$. Let P be a point on $\overline{BC}$ such that $AP + PD$ is the smallest possible value. If the perimeter of $\triangle ABC$ is 63, find the exact value of $(AP + PD)$.
18. Let $i = \sqrt{-1}$ and let k and w represent real numbers. If $(k + wi)(4 + i) = 33 - 13i$ then find the value of $(k + w)$.
19. Find the acute angle between the normal vectors of the following two planes:
 $3x + 2y + 5z + 4 = 0$ and $-2x - 3y + 6z - 2 = 0$
Round your answer to the nearest minute and then express your answer in degrees and minutes in the form: $k^\circ w'$.
20. Points A , B , C , and D are the vertices of a rectangle. Point E is in the interior of this rectangle. $AE = 8$, $EC = 21$. The lengths of $\overline{DE}$, $\overline{BE}$, and $\overline{BC}$ are integers. If $DE < BE$, find the largest possible value for the length of $\overline{BC}$.